9th Class • Punjab Board • PECTAA New Syllabus

Chapter 8: Magnetism

Complete solved exercise — fields, domains, magnetisation, solenoids, electromagnetic induction and circuit protection.

Chapter 8 • Complete official exercise • 8 MCQs + 7 short + 5 constructed + 6 comprehensive

What this chapter is about

- **Magnetic poles** occur in pairs: like poles repel and unlike poles attract. A magnet also attracts unmagnetised ferromagnetic material by induced magnetism.
- **Field lines** run N → S outside and S → N inside a magnet, forming non-intersecting closed loops. Greater line density indicates stronger field.
- **Domains** explain magnetisation. Soft iron aligns and realigns easily; steel and hard magnetic alloys retain alignment for permanent magnets.
- **Electromagnetism:** A current-carrying solenoid produces a bar-magnet-like field. Right-hand rule gives its poles; changing magnetic flux can induce current.

Compass rule: The north pole of a compass needle points along the local magnetic field. Viewed from a solenoid end, anticlockwise conventional current makes that end N; clockwise makes it S.

A. Multiple Choice Questions

1. Which one of the following is not a magnetic material?

(a) Cobalt (b) Iron
(c) Aluminium (d) Nickel

Answer: (c) Aluminium — Iron, cobalt and nickel are ferromagnetic. Aluminium is only weakly paramagnetic and is not attracted like an ordinary magnetic material.

2. Magnetic lines of force:

(a) are always directed in a straight line (b) cross one another
(c) enter into the north pole (d) enter into the south pole

Answer: (d) enter into the south pole — Outside a magnet, lines leave the north pole and enter the south pole. They form closed loops through the magnet and never cross.

3. Permanent magnets cannot be made by:

(a) soft iron (b) steel
(c) neodymium (d) alnico

Answer: (a) soft iron — Soft iron magnetises strongly but loses most magnetism when the external field is removed, making it suitable for temporary electromagnets rather than permanent magnets.

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4. Permanent magnets are used in:

(a) circuit breaker
(b) loudspeaker
(c) electric crane
(d) magnetic recording

Answer: (b) loudspeaker — A moving-coil loudspeaker uses a permanent magnet's field and a current-carrying coil. Circuit breakers and lifting cranes normally use electromagnets.

5. A common method used to magnetise a material is:

(a) stroking
(b) hitting
(c) heating
(d) placing inside a solenoid having A.C.

Answer: (a) stroking — Stroking with a permanent magnet helps align domains. Hitting, heating and an alternating field are common demagnetising methods.

6. A compass is placed above, below, left and right of a bar magnet with N on the left and S on the right. Which diagram has correct field directions?

(a) (a)
(b) (b)
(c) (c)
(d) (d)

Answer: (c) (c) — Outside field runs N → S: rightward above and below, outward left from the N end and leftward into the S end. This is diagram (c).

7. A steel rod is magnetised by double-touch stroking, moving the shown magnets from the centre outwards. Which polarity of AB is correct?

(a) A = N, B = S
(b) A = S, B = N
(c) A = S, B = S
(d) A = N, B = N

Answer: (a) A = N, B = S — The S pole stroking toward A leaves A north, and the N pole stroking toward B leaves B south. Thus A = N and B = S, option (a).

8. The best material to protect a device from external magnetic field is:

(a) wood
(b) plastic
(c) steel
(d) soft iron

Answer: (d) soft iron — High-permeability soft iron provides a low-reluctance path for magnetic flux around the shielded space. Wood and plastic do not provide magnetic shielding.

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B. Short Answer Questions

8.1. What are temporary and permanent magnets?

A **temporary magnet** shows magnetism mainly while an external magnetising field or current is present and loses it readily, as in soft iron. A **permanent magnet** retains substantial magnetisation after the external field is removed, as in steel, alnico or neodymium alloys.

8.2. Define magnetic field of a magnet.

The magnetic field is the region around a magnet or current in which another magnet, magnetic material or moving charge can experience a magnetic force. At each point it has magnitude and direction, represented by the vector B.

8.3. What are magnetic lines of force?

They are imagined continuous curves whose tangent gives magnetic-field direction and whose spacing represents strength. Outside a magnet they run from N to S and inside from S to N. They form closed loops and cannot intersect because field direction at a point is unique.

8.4. Name some uses of permanent magnets and electromagnets.

Permanent magnets	Electromagnets
Moving-coil loudspeakers and microphones	Electric bells and relays
Electric motors and generators	Circuit breakers and contactors
Moving-coil meters and compasses	Scrap-lifting cranes and magnetic separators
Magnetic catches and sensors	Telephone receivers and recording heads

8.5. What are magnetic domains?

Magnetic domains are tiny regions in a ferromagnetic material within which atomic magnetic moments are aligned in the same direction. An unmagnetised sample has randomly oriented domains that largely cancel; magnetisation aligns more domains and their boundaries shift, producing a strong net field.

8.6. Which type of magnetic field is formed by a current-carrying long coil?

A long current-carrying coil or **solenoid** produces a nearly uniform, straight and parallel field inside and a bar-magnet-like field outside. One end behaves as N and the other S; the right-hand grip rule relates current direction to field and poles.

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8.7. Differentiate between paramagnetic and diamagnetic materials.

Paramagnetic	Diamagnetic
Atomic orbital and spin magnetic effects do not completely cancel; atoms have small net moments.	Atomic magnetic effects cancel, leaving no permanent atomic magnetic moment.
Weakly align with and are attracted toward stronger field.	Induced magnetism opposes the applied field, causing very weak repulsion.
Lose alignment when field is removed; do not form permanent domain magnets.	Opposing induced effect disappears when field is removed.
Examples include aluminium and platinum.	Examples include copper, bismuth and water.

C. Constructed Response Questions

8.1. Two bar magnets are stored vertically in a wooden box. Their colours show left magnet N above S and right magnet S above N. Label the poles and identify the top and bottom objects P and Q.

Position	Left magnet	Right magnet
Top, touching P	N	S
Bottom, touching Q	S	N

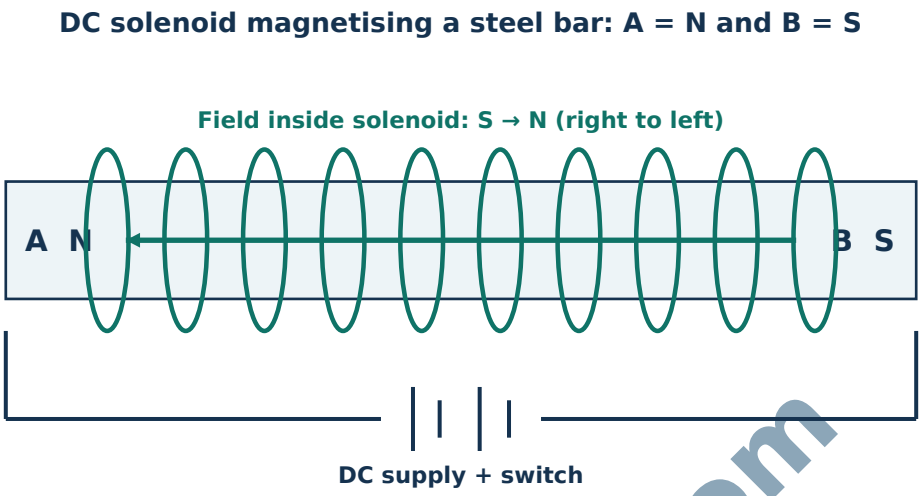
P and Q are soft-iron keepers, not permanent magnets. They bridge opposite poles and complete low-reluctance magnetic circuits, reducing self-demagnetisation during storage.

Induced polarity in P is S near the left N and N near the right S. In Q it is N near the left S and S near the right N.

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8.2. A steel bar is to be magnetised inside a solenoid so that end A (left) becomes N and end B (right) becomes S. Draw the circuit diagram and state the current rule.

Figure 1 — Steel bar inside a DC solenoid with the required polarity



Right-hand rule: viewed from A, the conventional current is anticlockwise for a north pole at A.

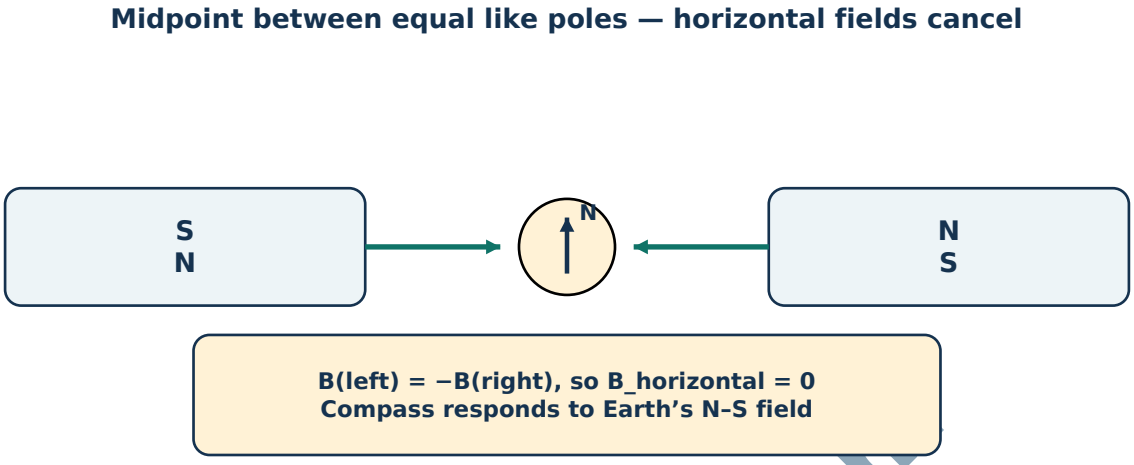
Use a **DC supply and switch**, not AC. The solenoid field inside points from B(S) toward A(N), aligning domains in the steel. Viewed from end A, conventional current must be **anticlockwise**; viewed from B it is clockwise.

Switching off leaves much of the alignment in steel. Soft iron would lose most of it and is used when a controllable temporary electromagnet is required.

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8.3. Two equal bar magnets lie end-to-end with a compass at the midpoint of the gap. The needle settles in the north–south direction, perpendicular to the magnet line. Label possible N and S poles and justify with field lines.

Figure 2 — One valid arrangement: equal N poles face the gap



The same cancellation occurs for S–S facing poles; orientation alone cannot distinguish N–N from S–S.

The facing ends must be **equal like poles**, either N–N or S–S. At the midpoint their equal horizontal fields cancel, so the compass responds to Earth's north–south magnetic field rather than to a resultant horizontal magnet field.

For the N–N case shown, the outer ends are both S. For an S–S facing pair, the outer ends are both N. The needle orientation alone cannot distinguish these two cases, so a unique N/S colour assignment is underdetermined unless another field test is supplied.

8.4. Electric current or moving electrons produce a magnetic field. Is the reverse process true—can a magnetic field give rise to electric current? If yes, give an example.

Yes, but the magnetic flux through a circuit must **change**. Electromagnetic induction produces an emf; a closed circuit then carries induced current. A steady magnetic field with no relative motion or flux change does not sustain an induced current.

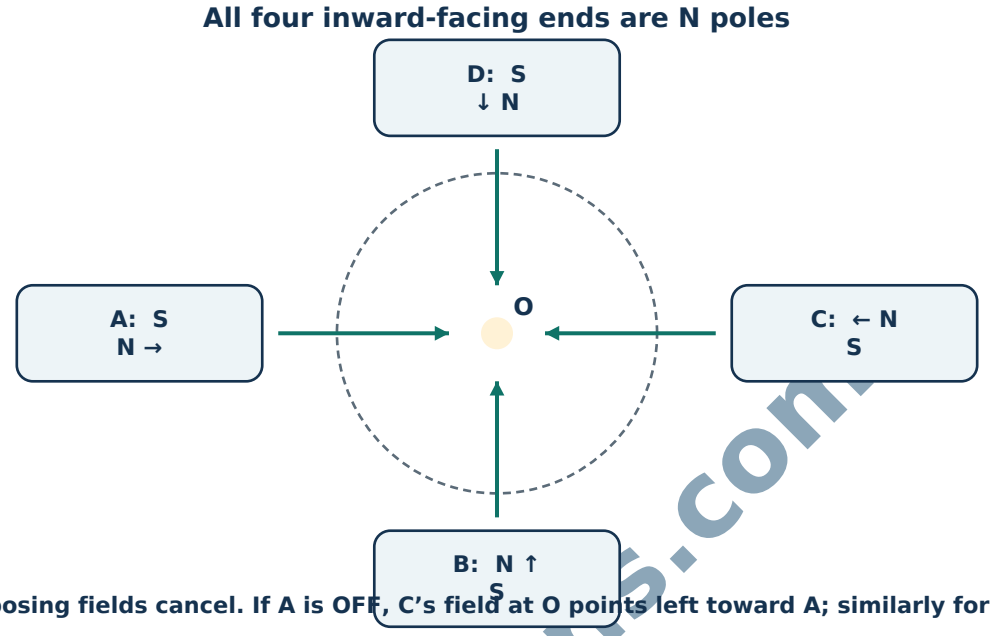
Example: push a bar magnet into a coil connected to a galvanometer. The pointer deflects while flux increases. Withdrawing the magnet reverses the deflection, and holding it stationary gives zero current.

A generator uses the same principle: a coil rotates in a magnetic field, continually changing flux and generating alternating emf. Mechanical energy is converted into electrical energy.

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8.5. Four identical solenoids A (left), B (bottom), C (right) and D (top) surround centre O. Equal currents are used. Choose current directions so that switching any one OFF leaves the net field at O directed toward that solenoid.

Figure 3 — Required poles and field directions at the centre



Equal opposing fields cancel. If A is OFF, C's field at O points left toward A; similarly for B, C or D.

Current is anticlockwise when each inward N end is viewed from O (right-hand rule).

Make **every inward-facing end a north pole**. A's right field points right and C's left field points left, so they cancel; B's upward and D's downward fields also cancel.

If A is switched OFF, C remains and points the field left toward A. If C is OFF, A points right toward C; the same logic applies to B and D. By the right-hand rule, current at each inward N end is anticlockwise when viewed from O.

D. Comprehensive Questions

8.1. How can you identify whether an object is a magnet or a magnetic material?

Use a known magnet and bring its **two poles separately** near each end of the object.

• **Repulsion occurs:** The object is definitely a magnet. Only like magnetic poles repel; an unmagnetised magnetic material cannot repel a permanent magnet.

• **Attraction occurs with both poles:** The object is an unmagnetised magnetic material such as iron or steel, or possibly a magnet with a particular orientation. Further orientation testing is needed.

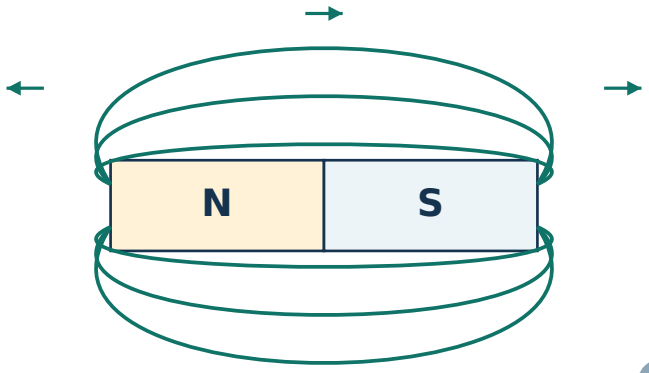
• **No noticeable attraction:** It is nonmagnetic under ordinary conditions, such as wood, plastic or aluminium.

The definitive test is **repulsion**, because attraction alone can result from induced magnetism. Freely suspending a magnet so that it aligns approximately north–south is a useful supporting test.

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8.2. Describe the strength of a magnetic field in terms of magnetic lines of force. Explain with diagrams.

Bar-magnet field — outside the magnet, lines run from N to S



Closer lines near the poles indicate stronger field; wider spacing indicates weaker field.

A compass needle’s north pole follows the local tangent; field lines never cross.

Magnetic field strength is represented by the number of field lines passing through a unit area placed perpendicular to them. **Closer lines mean stronger field**; lines farther apart mean weaker field.

Around a bar magnet, lines are dense near the poles where force is strongest and spread out farther away. In a uniform field, such as inside a long solenoid, parallel equally spaced lines represent constant magnitude and direction.

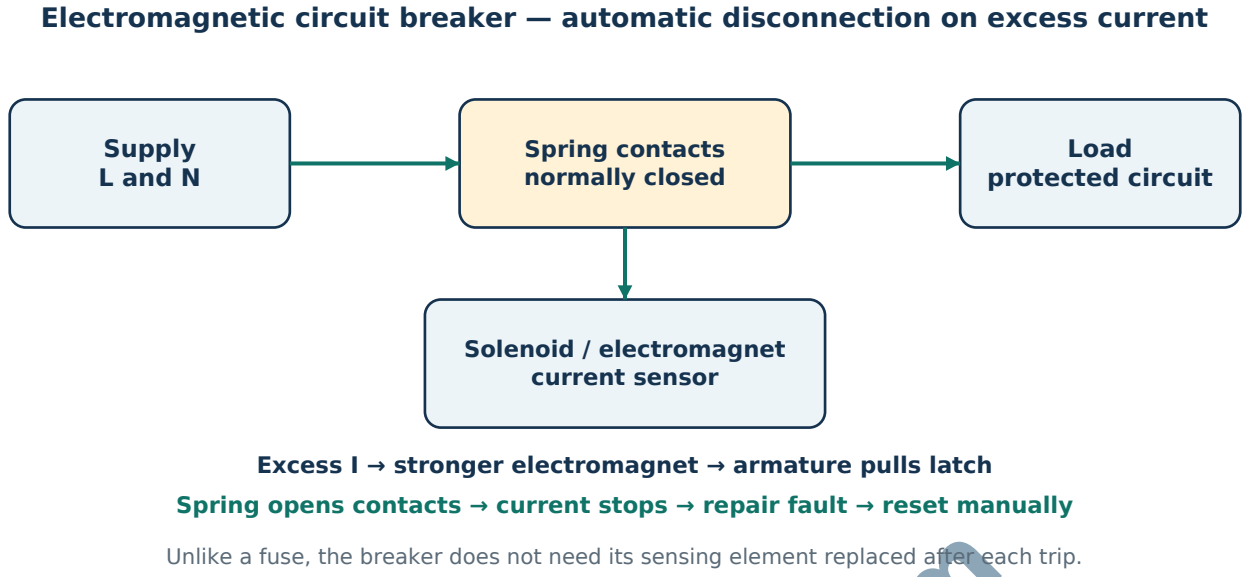
Lines never cross: two crossing tangents would imply two field directions at one point. They are closed loops, leaving N and entering S outside and continuing S to N inside.

8.3. What is a circuit breaker? Describe its working with the help of a diagram.

A **circuit breaker** is an automatic safety switch that disconnects a circuit when current exceeds a safe value because of overload or short circuit. It protects wiring and appliances from overheating and fire.

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Figure 5 — Simplified electromagnetic trip mechanism



1. During normal operation, spring-loaded contacts are closed and load current flows through a sensing coil.

2. Excess current makes the electromagnet strong enough to pull an armature and release the latch.

3. The spring rapidly opens the contacts and interrupts current.

4. After the fault is removed, the breaker is reset manually; it does not need replacement like a blown fuse.

Real breakers may use thermal, magnetic or combined trip mechanisms and must have the correct current rating.

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8.4. “A magnet attracts only a magnet.” Explain the statement.

The statement needs correction. A magnet attracts **magnetic materials**, whether already magnetised or not; it does not attract every kind of material. Iron, cobalt, nickel and suitable alloys respond strongly.

When an unmagnetised iron object enters the field, its domains align by induction. The near side acquires opposite polarity to the magnet’s near pole and the far side like polarity. Attraction dominates because the opposite induced pole is closer to the magnet.

Between two magnets, unlike poles attract and like poles repel. Thus attraction does not prove that an object was already a magnet; only repulsion provides that definite test.

8.5. Differentiate between paramagnetic, diamagnetic and ferromagnetic materials with reference to domain theory.

Type	Atomic / domain behaviour	Response to external field
Paramagnetic	Orbital and spin effects support a small net atomic moment, but strong ordered domains are absent.	Moments align weakly with field; very weak attraction and no lasting magnetisation.
Diamagnetic	Orbital and spin contributions cancel to zero net atomic moment; no magnetic domains.	Field induces a small opposing moment, causing very weak repulsion.
Ferromagnetic	Large groups of atoms have parallel moments within domains; domain directions can be random initially.	Domains grow and rotate into strong alignment; high permeability and possible retained magnetisation.

Ferromagnetism disappears above a material-specific Curie temperature because thermal agitation destroys domain order. Paramagnetism and diamagnetism are much weaker effects.

8.6. Why are ferromagnetic materials suitable for making magnets?

Their domain structure allows strong cooperative alignment of many atomic magnetic moments. An external field moves domain walls and rotates domains, producing magnetisation far stronger than in para- or diamagnetic substances.

• **Soft ferromagnetic materials** such as soft iron align and lose alignment easily, ideal for electromagnet cores, relays and transformer cores.

• **Hard ferromagnetic materials** such as steel, alnico and neodymium alloys resist domain realignment, so they retain magnetisation and serve as permanent magnets.

• **High permeability** concentrates magnetic flux and produces strong fields and forces with practical sizes and currents.

The choice therefore depends on whether retained magnetism or easy switching is required.

E. More Important Questions (from the text)

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1. Compare soft iron and steel as magnetic materials.

Property	Soft iron	Steel
Magnetisation	Quick and strong in external field.	Harder to magnetise but can become strong.
Retentivity	Low; loses most magnetism when field is removed.	High; retains magnetisation for a long time.
Domain realignment	Easy; low coercivity.	Difficult; higher coercivity.
Typical use	Electromagnet and transformer cores, magnetic shielding.	Permanent magnets for speakers, meters and motors.

2. How can a steel bar be magnetised and demagnetised?

Magnetising methods	Demagnetising methods
Single- or double-touch stroking with a permanent magnet in one repeated direction.	Hammering or dropping while oriented away from Earth’s field, which disturbs domains.
Placing inside a solenoid carrying strong direct current.	Heating above the Curie temperature and cooling without an aligning field.
Holding in a strong external magnetic field so domains align.	Withdrawing gradually from a solenoid carrying alternating current, which repeatedly reverses and reduces alignment.

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Points to Remember

Poles: like repel, unlike attract; isolated N or S poles do not occur.

Repulsion is the definitive test that an object is already magnetised.

Field lines: N → S outside, S → N inside, closed loops and no crossings.

Strength: closer perpendicular line density means stronger field.

Compass: its north pole points along local field direction.

Domains: random orientations cancel; aligned domains produce net magnetisation.

Soft iron is temporary and easily switched; steel and hard alloys retain magnetism.

Solenoid: nearly uniform internal field; anticlockwise current at a viewed end gives N.

Shielding: high-permeability soft iron guides flux around a protected device.

Induction: changing magnetic flux, not a steady field alone, induces emf and current.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	c	2	d	3	a	4	b	5	a
6	c	7	a	8	d				

How to attempt this chapter in the paper

• Draw field arrows before labelling compass needles; outside a magnet always check N → S.
• When a question supplies only cancellation at a midpoint, do not assume whether facing like poles are N or S without more evidence.
• Use DC to magnetise with a solenoid and AC with gradual withdrawal to demagnetise.
• Do not use attraction alone to identify a permanent magnet; induced magnetic materials also attract.
• For solenoid arrangements, cancel opposing pairs first and then inspect the remaining field.
• Distinguish retentivity from permeability: easy magnetisation does not necessarily mean permanent magnetisation.